

Complete the table by rewriting each multiplication expression using the distributive property, modeling with area, and finding the product.

Multiplication Expression	Distributive Property	Area Model	Product
1. 41×8	$8(40 + 1)$ $= 320 + 8$ $= 328$	<u>8</u> 40 320 + 1 8	328
2. 6×89	$6(80 + 9)$ $= 480 + 54$ $= 534$	<u>6</u> 80 480 + 9 54	534
3. 96×3	$3(90 + 6)$ $= 270 + 18$ $= 288$	<u>3</u> 90 270 + 6 18	288

For each expression, generate an equivalent numeric expression using the distributive property and then find the value.

4. 9×42

$$\begin{aligned}
 &9(40 + 2) \\
 &= 9 \times 40 + 9 \times 2 \\
 &= 360 + 18 \\
 &= 378
 \end{aligned}$$

5. $28 \cdot 7$

$$\begin{aligned}
 &7(20 + 8) \\
 &= 7 \times 20 + 7 \times 8 \\
 &= 140 + 56 \\
 &= 196
 \end{aligned}$$

6. 5×64

$$\begin{aligned}
 &5(60 + 4) \\
 &= 5 \times 60 + 5 \times 4 \\
 &= 300 + 20 \\
 &= 320
 \end{aligned}$$

Rewrite an equivalent expression for each sum using the distributive property with the greatest common factor (GCF).

7. $90 + 75$

$15(6 + 5)$

8. $208 + 336$

$16(13 + 21)$

9. Generate two different equivalent expressions for the sum $340 + 220$ by rewriting it using the distributive property with two different common factors.

Answers may vary.

$2(170 + 110)$

$10(34 + 22)$

10. Jayden used the distributive property to write an expression equivalent to $48 + 72$. His expression was $4(8 + 18)$. Is Jayden correct? If yes, explain how he determined his expression. If not, explain Jayden's error and write a correct equivalent expression.

He is not correct because $4 \times 8 = 32$ not 48. It should be $4(12 + 18)$.